## ECM validation – Summary & Equations

Duration: 34999.16 s    dt ≈ 1.0000 s    n = 35000

RMSE = 0.030662 V    MAE = 0.023560 V    Max|err| = 0.172098 V @ 29900.139 s    Bias = 0.013723 V

## Model & error definitions

$$V_{err}(t) = V_{meas}(t) - V_{ter}(t)$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} V_{err,i}^2}$$

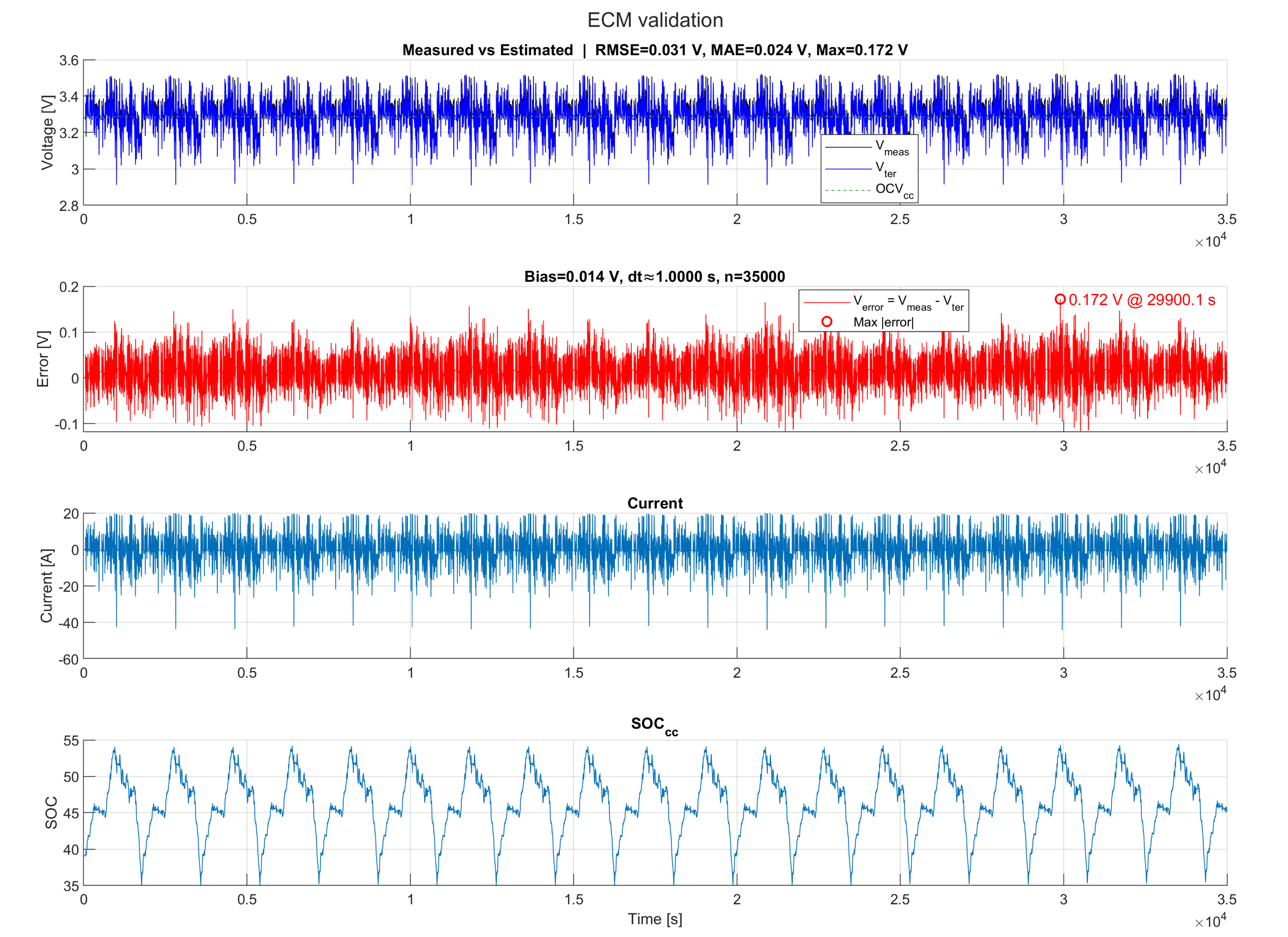
$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |V_{err,i}|$$

$$\mathrm{Bias} = \frac{1}{n}\sum_{i=1}^{n} V_{err,i}$$

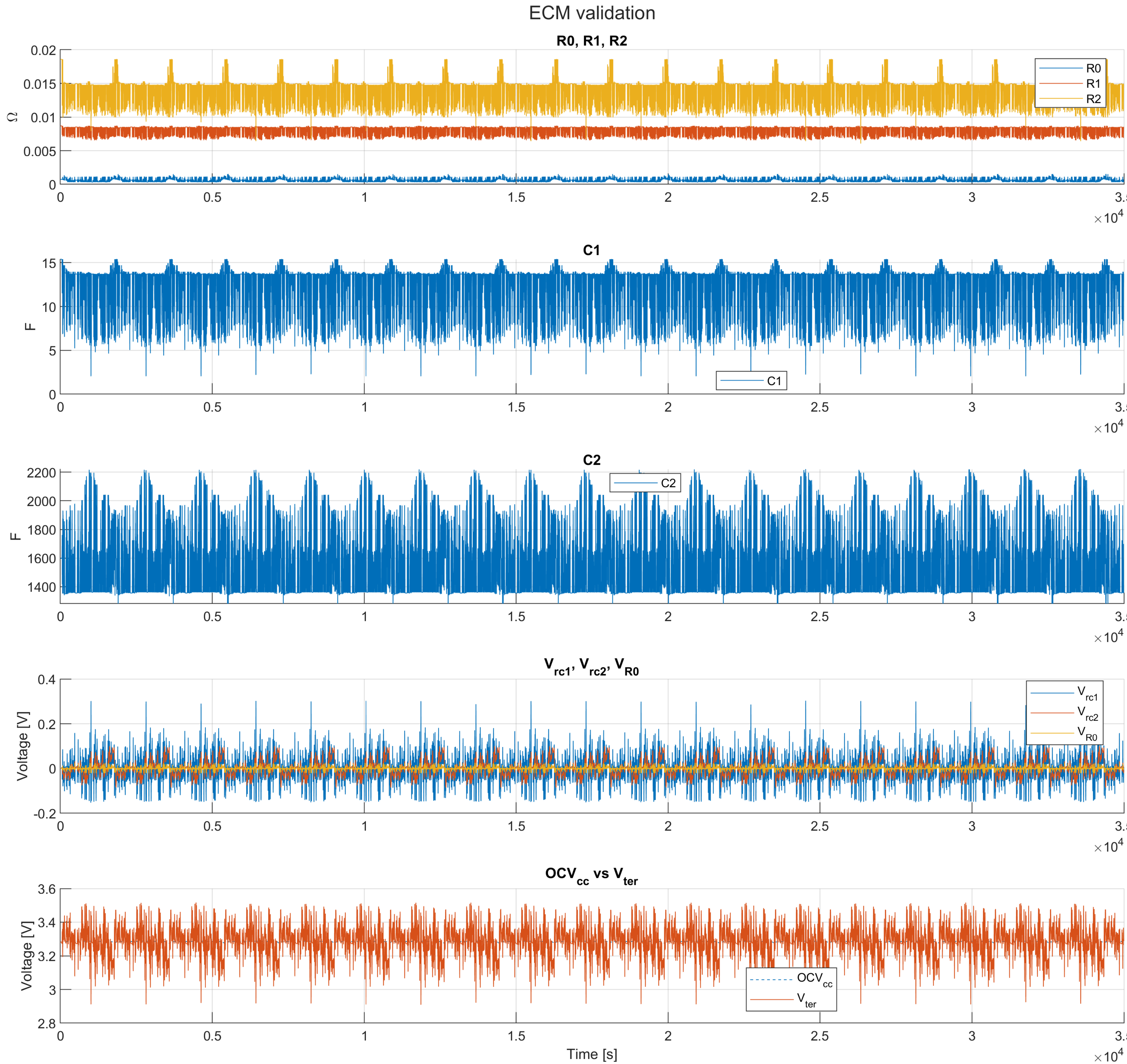
$$V_{ter}(t) = OCV(SOC(t)) - R_0(t)\, I(t) - V_{rc1}(t) - V_{rc2}(t)$$

$OCV_{cc}$: yes    R0: yes    R1: yes    R2: yes    C1: yes    C2: yes

$V_{rc1}$: yes    $V_{rc2}$: yes    $V_{R0}$: yes



ECM validation
Measured vs Estimated | RMSE=0.031 V, MAE=0.024 V, Max=0.172 V
Voltage [V]
V_meas
V_ter
OCV_cc
Bias=0.014 V, dt≈1.0000 s, n=35000
Error [V]
V_error = V_meas - V_ter
Max |error|
0.172 V @ 29900.1 s
Current
Current [A]
SOC_cc
SOC
Time [s]
×10^4


ECM validation
R0, R1, R2
Ω
R0
R1
R2
C1
F
C2
$V_{rc1}$, $V_{rc2}$, $V_{R0}$
Voltage [V]
$OCV_{cc}$ vs $V_{ter}$
$OCV_{cc}$
$V_{ter}$
Time [s]
$\times 10^4$